

N-acetylcysteine in acute hepatic failure (non-paracetamol-induced).

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Acute liver failure is a serious condition associated with poor prognosis. It may be associated with changes in systemic hemodynamics, i.e., tissue hypoxia, which contributes to multiple-organ failure. Recent studies have shown that N-acetylcysteine administered to patients with fulminant hepatic failure (paracetamol-induced) increases oxygen delivery and improves survival. The aim of this pilot study was to evaluate N-acetylcysteine administration to patients with non-paracetamol-induced acute liver failure and assess its effect on the clinical course and outcome. N-acetylcysteine was administered at presentation to 7 patients with non-paracetamol-induced acute liver failure. Patients were followed for changes in clinical parameters (grade of encephalopathy), coagulation factors, biochemical parameters and outcome. Clinically, 3 patients who initially had grade O/II encephalopathy, did not progress, and have fully recovered. The mean peak prothrombin time, serum factor V, aspartate aminotransferase and alanine aminotransferase levels, all significantly improved. Four patients (57%) have recovered fully (1 patient, although fully recovered, died later from an unrelated cause). Two patients required orthotopic liver transplantation and 1 patient died. N-acetylcysteine administration may have prevented progression to grade III/IV encephalopathy and improved serum coagulation factors. This may account for its beneficial effect on survival in patients who had poor prognostic criteria at base-line. No side effects of the drug were noted. This study suggests that N-acetylcysteine administration should be considered in all patients with acute liver failure.

N-acetylcysteine for non-paracetamol (acetaminophen)-related acute liver failure.

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Acute liver failure is a rare and serious disease. Acute liver failure may be paracetamol-induced or non-paracetamol-induced. Acute liver failure not caused by paracetamol (acetaminophen) has a poor prognosis with limited treatment options. N-acetylcysteine has been successful in treating paracetamol-induced acute liver failure and reduces the risk of needing to undergo liver transplantation. Recent randomised clinical trials have explored whether the benefit can be extrapolated to treat non-paracetamol-related acute liver failure. The American Association for the Study of Liver Diseases (AASLD) 2011 guideline suggested that N-acetylcysteine could improve spontaneous survival when given during early encephalopathy stages for patients with non-paracetamol-related acute liver failure. To assess the benefits and harms of N-acetylcysteine compared with placebo or no N-acetylcysteine, as an adjunct to usual care, in people with non-paracetamol-related acute liver failure. We searched the Cochrane Hepato-Biliary Group Controlled Trials Register (searched 25 June 2020), Cochrane Central Register of Controlled Trials (CENTRAL; 2020, Issue 6) in The Cochrane Library, MEDLINE Ovid (1946 to 25 June 2020), Embase Ovid (1974 to 25 June 2020), Latin American and Caribbean Health Science Information database (LILACS) (1982 to 25 June 2020), Science Citation Index Expanded (1900 to 25 June 2020), and Conference Proceedings Citation Index - Science (1990 to 25 June 2020). We included randomised clinical trials that compared N-acetylcysteine at any dose or route with placebo or no intervention in participants with non-paracetamol-induced acute liver failure. We used standard methodological procedures as described in the Cochrane Handbook for Systematic Reviews of Interventions. We conducted meta-analyses and presented results using risk ratios (RR) with 95% confidence intervals (CIs). We quantified statistical heterogeneity by calculating I². We assessed bias using the Cochrane risk of bias tool and determined the certainty of the evidence using the GRADE approach. We included two randomised clinical trials: one with 183 adults and one with 174 children

(birth through age 17 years). We classified both trials at overall high risk of bias. One unregistered study in adults is awaiting classification while we are awaiting responses from study authors for details on trial methodology (e.g. randomisation processes). We did not meta-analyse all-cause mortality because of significant clinical heterogeneity in the two trials. For all-cause mortality at 21 days between adults receiving N-acetylcysteine versus placebo, there was inconclusive evidence of effect (N-acetylcysteine 24/81 (29.6%) versus placebo 31/92 (33.7%); RR 0.88, 95% CI 0.57 to 1.37; low certainty evidence). The certainty of the evidence was low due to risk of bias and imprecision. Similarly, for all-cause mortality at one year between children receiving N-acetylcysteine versus placebo, there was inconclusive evidence of effect (25/92 (27.2%) versus 17/92 (18.5%); RR 1.47, 95% CI 0.85 to 2.53; low certainty evidence). We downgraded the certainty of evidence due to very serious imprecision. We did not meta-analyse serious adverse events and liver transplantation at one year due to incomplete reporting and clinical heterogeneity. For liver transplantation at 21 days in the trial with adults, there was inconclusive evidence of effect (RR 0.72, 95% CI 0.49 to 1.06; low certainty evidence). We downgraded the certainty of the evidence due to serious risk of bias and imprecision. For liver transplantation at one year in the trial with children, there was inconclusive evidence of effect (RR 1.23, 95% CI 0.84 to 1.81; low certainty of evidence). We downgraded the certainty of the evidence due to very serious imprecision. There was inconclusive evidence of effect on serious adverse events in the trial with children (RR 1.25, 95% CI 0.35 to 4.51; low certainty evidence). We downgraded the certainty of the evidence due to very serious imprecision. We did not meta-analyse non-serious adverse events due to clinical heterogeneity. There was inconclusive evidence of effect on non-serious adverse events in adults (RR 1.07, 95% CI 0.79 to 1.45; 173 participants; low certainty of evidence) and children (RR 1.19, 95% CI 0.62 to 2.16; 184 participants; low certainty of evidence). None of the trials reported outcomes of proportion of participants with resolution of encephalopathy and coagulopathy or health-related quality of life. The National Institute of Health in the United States funded both trials through grants. One of the trials received additional funding from two hospital foundations' grants. Pharmaceutical companies provided the study drug and matching placebo, but they did not have input into study design nor involvement in analysis. The available evidence is inconclusive regarding the effect of N-acetylcysteine compared with placebo or no N-acetylcysteine, as an adjunct to usual care, on mortality or transplant rate in non-paracetamol-induced acute liver failure. Current evidence does not support the guideline suggestion to use N-acetylcysteine in adults with non-paracetamol-related acute liver failure, nor the rising use observed in clinical practice. The uncertainty based on current scanty evidence warrants additional randomised clinical trials with non-paracetamol-related acute liver failure evaluating N-acetylcysteine versus placebo, as well as investigations to identify predictors of response and the optimal N-acetylcysteine dose and duration.

Analysis of the Australia and New Zealand referral criteria for transfer to a liver unit for paracetamol overdose.

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Paracetamol overdose is common and can lead to fulminant hepatic failure. In cases that are not improving with standard medical therapy with N-acetylcysteine, some patients may require liver transplant. The Australia and New Zealand (ANZ) referral criteria for transfer to a liver unit have not been extensively studied for its predictive value. The aim of this study was to evaluate the ANZ referral criteria for predicting mortality in paracetamol overdose. This study involves a retrospective analysis of patients who developed hepatotoxicity post-paracetamol overdose presenting to an Australian health service with a liver transplant unit between 2010 and 2019 and were treated with N-acetylcysteine. The primary

outcome was death or transplant. Out of 983 paracetamol overdose presentations, 81 (8.2%) cases developed hepatotoxicity. Of these, 17 cases (21%) met the composite endpoint of death or transplant. The ANZ referral criteria is highly sensitive at predicting the primary endpoint of death or transplant at time of referral 100% (95% confidence interval 81-100) but had low specificity at 30% (95% confidence interval 19-42). The ANZ referral criteria were highly sensitive for predicting the outcome of mortality and transplant. This is important for screening patients who may become unstable and difficult to transfer at a later stage of their admission.

Improved outcome of paracetamol-induced fulminant hepatic failure by late administration of acetylcysteine.

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The influence of acetylcysteine, administered at presentation to hospital, on the subsequent clinical course of 100 patients who developed paracetamol-induced fulminant hepatic failure was analysed retrospectively. Mortality was 37% in patients who received acetylcysteine 10-36 h after the overdose, compared with 58% in patients not given the antidote. In patients given acetylcysteine, progression to grade III/IV coma was significantly less common than in those who did not receive the antidote (51% vs 75%), although the median peak prothrombin time was similar for both groups. Whether the beneficial effect is related to replenishment of glutathione stores or a consequence of another hepatic protective mechanism of acetylcysteine requires further study.

Safety of late acetylcysteine treatment in paracetamol poisoning.

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Twenty patients who had taken overdoses of paracetamol were treated with acetylcysteine between 12 and 24 hours after the incident. Although 19 patients had plasma paracetamol concentrations greater than those associated with a 90% risk of moderate to severe liver damage, this complication occurred in only seven (35%) individuals. No patient developed hepatic encephalopathy or acute renal failure and all recovered without sequelae. We conclude that acetylcysteine administration up to 24 hours following paracetamol overdose is not dangerous and may prevent further liver damage.

[Treatment of paracetamol poisoning. An indication for liver transplantation?].

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Development of metabolic acidosis (pH 100s) and oliguric renal failure are associated with a poor prognosis in paracetamol-induced fulminant liver failure. It is important to administer N-acetylcysteine as soon as possible after the overdose, but N-acetyl-cysteine also seems to improve survival when given 36-80h following ingestion. Liver transplantation has been performed in some patients with paracetamol-induced fulminant liver failure, but convincing evidence that transplantation improves survival in this group of patients is still lacking. We discuss the difficulties met in deciding if and when to perform liver transplantation. Renal failure may develop some days after paracetamol poisoning, even in

the absence of severe liver damage, and haemofiltration and haemodialysis may be necessary.

Intravenous N-acetylcysteine improves transplant-free survival in early stage non-acetaminophen acute liver failure.

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N-acetylcysteine (NAC), an antidote for acetaminophen poisoning, might benefit patients with non-acetaminophen-related acute liver failure. In a prospective, double-blind trial, acute liver failure patients without clinical or historical evidence of acetaminophen overdose were stratified by site and coma grade and assigned randomly to groups that were given NAC or placebo (dextrose) infusion for 72 hours. The primary outcome was overall survival at 3 weeks. Secondary outcomes included transplant-free survival and rate of transplantation. A total of 173 patients received NAC ($n = 81$) or placebo ($n = 92$). Overall survival at 3 weeks was 70% for patients given NAC and 66% for patients given placebo (1-sided $P = .283$). Transplant-free survival was significantly better for NAC patients (40%) than for those given placebo (27%; 1-sided $P = .043$). The benefits of transplant-free survival were confined to the 114 patients with coma grades I-II who received NAC (52% compared with 30% for placebo; 1-sided $P = .010$); transplant-free survival for the 59 patients with coma grades III-IV was 9% in those given NAC and 22% in those given placebo (1-sided $P = .912$). The transplantation rate was lower in the NAC group but was not significantly different between groups (32% vs 45%; $P = .093$). Intravenous NAC generally was well tolerated; only nausea and vomiting occurred significantly more frequently in the NAC group (14% vs 4%; $P = .031$). Intravenous NAC improves transplant-free survival in patients with early stage non-acetaminophen-related acute liver failure. Patients with advanced coma grades do not benefit from NAC and typically require emergency liver transplantation.

Intravenous acetylcysteine for indications other than acetaminophen overdose.

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The use of intravenous acetylcysteine for off-label indications, specifically non-acetaminophen-induced acute liver failure (NAI-ALF), severe alcoholic hepatitis, and contrast-induced nephropathy (CIN), is reviewed. I.V. acetylcysteine is most often used as an antidote for acetaminophen overdose due to its ability to increase levels of glutathione; however, it is also used to treat NAI-ALF and severe alcoholic hepatitis and to prevent CIN. Although the i.v. and oral formulations of acetylcysteine have been evaluated for these indications, most studies have examined the i.v. form. I.V. acetylcysteine is used in the treatment of NAI-ALF to improve oxygenation to the liver. One large randomized trial of 173 adults with NAI-ALF from any etiology and of any grade encephalopathy demonstrated overall improvement in transplant-free survival, particularly for patients with low-grade encephalopathy, though overall survival was not improved. When used to treat severe alcoholic hepatitis, i.v. acetylcysteine serves as an antioxidant and glutathione source. A trial of 174 patients with severe alcoholic hepatitis revealed that patients had 28-day survival benefit when treated with acetylcysteine; improvement in patients with hepatorenal syndrome was also noted. When used for the prevention of CIN, i.v. acetylcysteine provides antioxidants and vasodilation. The benefit for this indication is limited to surrogate markers such as serum creatinine and in patients with multiple risk factors for the development of CIN. Data regarding the use of i.v. acetylcysteine for the treatment of NAI-ALF and severe alcoholic hepatitis and in the

prevention of CIN are inconclusive, though some evidence supports its use in certain populations.

A prospective evaluation of shortened course oral N-acetylcysteine for the treatment of acute acetaminophen poisoning.

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Treatment with a shortened duration of oral N-acetylcysteine (20 to 48 hours) after acute acetaminophen poisoning is effective in the prevention of subsequent hepatic failure and death when administered to individuals meeting appropriate laboratory criteria. Individuals with a potentially toxic acetaminophen ingestion according to serum acetaminophen levels were identified prospectively using a large statewide poison control system database throughout a 12-month period. N-acetylcysteine was administered for a minimum of 6 doses (20 hours), after which laboratory studies were obtained. Discontinuation of N-acetylcysteine was recommended by the poison center when 2 criteria were met: serum acetaminophen was undetectable (<10 microg/mL) and liver test results were normal (serum aminotransferase, international normalized ratio). A follow-up questionnaire was administered to individuals treated with N-acetylcysteine for 48 hours or less to ascertain the presence of symptoms consistent with progressive hepatotoxicity. Of 205 acutely poisoned individuals treated with N-acetylcysteine for 48 hours or less, 195 were successfully contacted after discharge, and 187 of 195 (95.9%) reported no symptoms consistent with hepatic failure. Eight individuals (4.1%) reported abdominal pain or vomiting; however, none received further N-acetylcysteine treatment or additional hospitalization. A shortened duration of treatment with N-acetylcysteine (20 to 48 hours) may be an effective treatment option in individuals considered to be at no further risk of developing liver toxicity according to the fulfillment of appropriate laboratory criteria before N-acetylcysteine discontinuation.

The Use and Potential Benefits of N-Acetylcysteine in Non-Acetaminophen Acute Liver Failure: An Etiology-Based Review.

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Acute liver failure represents a life-threatening organ dysfunction with high mortality rates and an urgent need for liver transplantation. The etiology of the disease varies widely depending on various socio-economic factors and is represented mainly by paracetamol overdose and other drug-induced forms of liver dysfunction in the developed world and by viral hepatitis and mushroom poisoning in less developed countries. Current medical care constitutes either specific antidotes or supportive measures to ensure spontaneous recovery. Although it has been proven to have beneficial effects in paracetamol-induced liver failure, N-acetylcysteine is widely used for all forms of acute liver failure. Despite this, few well-designed studies have been conducted on the assessment of the potential benefits, dose regimens, or route of administration of N-acetylcysteine in non-acetaminophen liver failure. This review aims to summarize the current evidence behind the use of this drug in different forms of liver failure.
